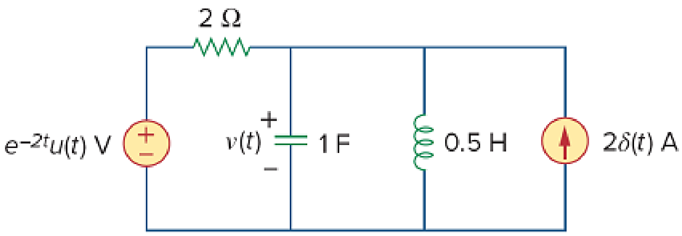
Chapter 18, Problem 41P

Problem

Determine the Fourier transform of $v(t)$ in the circuit shown in Fig. 18.43.

Figure 18.43



Step-by-step solution

Step 1 of 2

Refer to Figure 18.43 in the textbook.

The Fourier transform of voltage source is,

$$F\left(e^{-2t}u(t)\right)=\frac{1}{2+j\omega}$$

The Fourier transform of current source is,

$$F\left(2\delta(t)\right)=2$$

Draw the circuit in frequency domain.

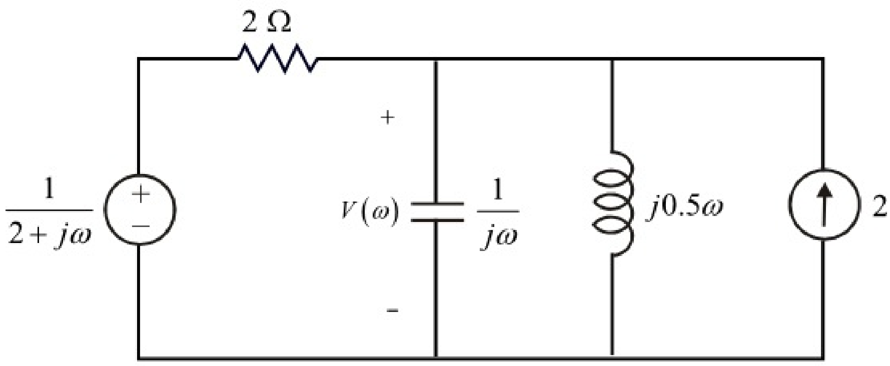


Figure 1

Step 2 of 2

Write the Kirchhoff's current law at node $V(\omega)$.

$$\frac{V(\omega)-\frac{1}{2+j\omega}}{2}+\frac{V(\omega)}{\frac{1}{j\omega}}+\frac{V(\omega)}{j0.5\omega}-2=0$$

$$\frac{V(\omega)-\frac{1}{2+j\omega}}{2}+j\omega V(\omega)+\frac{V(\omega)}{j0.5\omega}-2=0$$

$$V(\omega)\left(\frac{1}{2}+j\omega+\frac{1}{j0.5\omega}\right)=\frac{1}{2(2+j\omega)}+2$$

$$V(\omega)\left(\frac{j0.5\omega-\omega^2+2}{j\omega}\right)=\frac{1+8+4j\omega}{2(2+j\omega)}$$

$$V(\omega)=\frac{(9+4j\omega)(j\omega)}{2(2+j\omega)(j0.5\omega-\omega^2+2)}$$

$$V(\omega)=\frac{2j\omega(4.5+2j\omega)}{(2+j\omega)(4-2\omega^2+j\omega)}$$

Therefore, the Fourier transform of $v(t)$ is $\boxed{V(\omega)=\frac{2j\omega(4.5+2j\omega)}{(2+j\omega)(4-2\omega^2+j\omega)}}$.